



Deep Learning PROJECT SET491

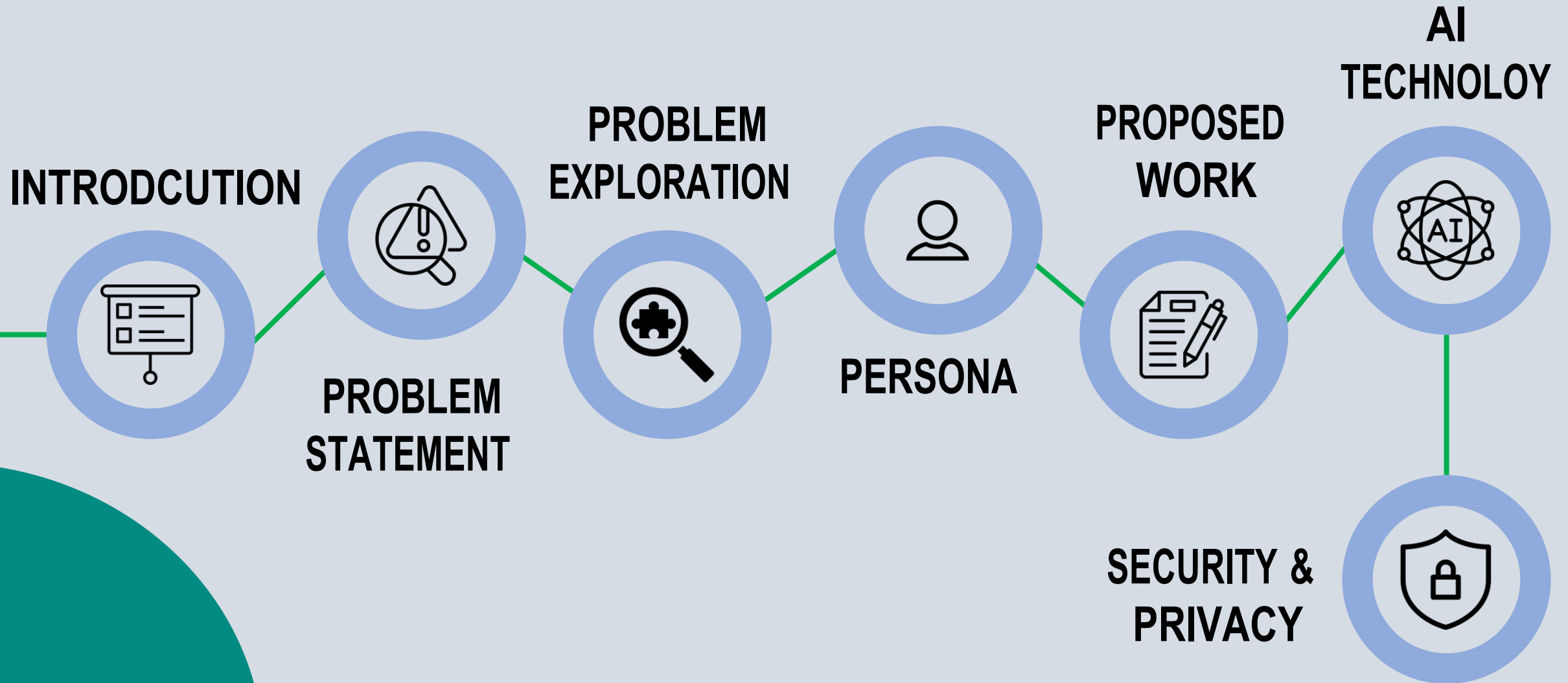
SUPERVISED BY:

DR. HOSSAM HAWASH

AETHEA



OUTLINES



INTRODUCTION



1

BLOOD TEST
ANALYSIS



2

MEDICINE
GUIDANCE



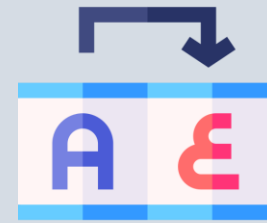
3

NUTRITION
PLANNING



4

ONLINE
CONSULTATION



5

ARABIC-FIRST
DESIGN

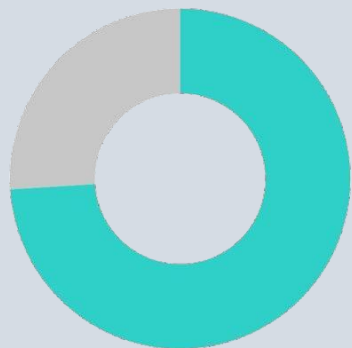


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DATA PRIVACY
AND SECURITY

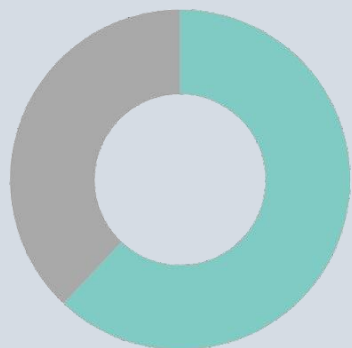
DIFFICULTY TO READ MEDICAL SCANS

74%



Overall Adult literacy rate

62% ↑



Adult Illiteracy
in rural areas

Source: World Bank Education Statistics, UNESCO data



DIFFICULTY TO READ MEDICAL SCANS



74.9%



Of medically-educated Egyptians studied at Arabic-medium schools

44.8%



Of medically-educated Egyptians studied at Arabic-medium schools

Source: WHO EMRO study on language barriers in medical education (2023)

DIFFICULTY TO READ MEDICAL SCANS

68.6%



of Egyptian patients over 60 years have
LIMITED health literacy

75%



of elderly
caregivers have
INADEQUATE health literacy

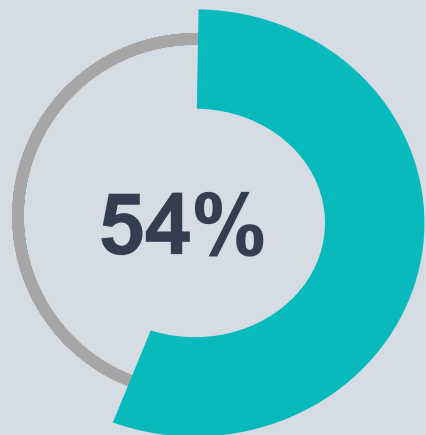
Source: Health literacy studies, Egyptian hospitals
Geriatric health study, Al-Azhar & Egyptian
universities, 2022-2023



LACK OF EXPERTISE IN RURAL AREAS

Rural Population in Egypt

approximately 60+
million people



of Egypt's total population
lives in rural areas

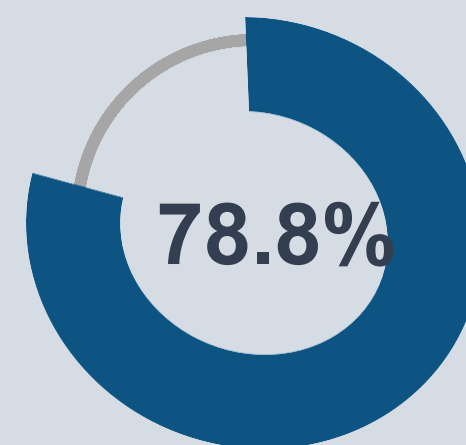
Rural vs Urban Healthcare Complaints

Urban Areas



reported health complaints

Rural Areas



reported health complaints

LACK OF EXPERTISE IN RURAL AREAS

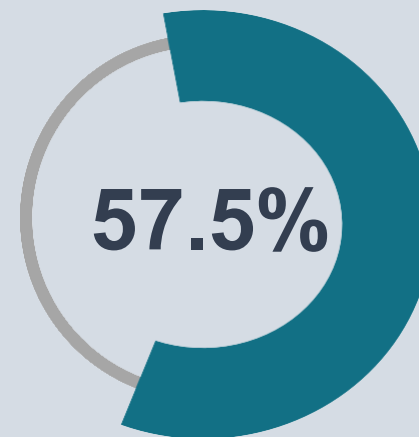
Doctor-to-Population Ratio

Approximately 1 doctor per
12,000 people in Egypt



Compared to 1 per
500-1000 in developed
countries)

Healthcare Access - Cannot Receive Treatment



of people in Egypt were unable to
receive treatment when they needed it

DOCTOR OVERCONFIDENCE

- 1 Quick diagnoses based on doctor confidence
- 2 Doctors may skip key diagnostic steps
- 3 Risk of misdiagnosis from cognitive bias

SLOW DIAGNOSIS

1

Emergencies require fast action

2

Remote areas lack immediate medical support

3

Delays due to scarcity of doctors



DIAGNOSTIC ERRORS

- 1 Inexperienced or unqualified doctors
- 2 High risk for serious diseases like cancer
- 3 Wrong diagnosis can endanger lives



SCATTERED MEDICAL SERVICES

1

Healthcare services are fragmented

2

Multiple platforms needed for basic care

3

Patients struggle to manage health tasks especially elderly people



BIAS CONCERNS

1

Discrimination in care based on identity

2

Insurance or public hospital bias

3

Need for fair and equal treatment



LACK OF MEDICAL CONFIDENCE

1

Fresh graduates lack clinical experience

2

Uncertainty in patient care decisions

3

Risks from inadequate training



SLOW HEALTHCARE ACCESS

- 1 Queue jumping and bribery disrupt fairness
- 2 No-shows hinder treatment
- 3 Long waits for clinical appointments

SELF-MEDICATION MISCONCEPTIONS

- 1 Widespread self-medication habits
- 2 Pills taken without consulting doctors
- 3 Risks from sharing or reusing prescriptions



OCCUPATION

Works at a busy community pharmacy



TECH LEVEL

Moderate, Uses Pharmacy management Software



NEEDS

Alerts to reduce errors, ensuring patient safety & drug clarity

Doctor Persona Ibrahim 35



KEY PAIN POINTS

1. Handwriting on prescriptions is often unclear
2. Patients sometimes misread dosage instructions
3. Cannot always check complex medicine interactions



OCCUPATION

A retired school teacher
lives in a small town



TECH LEVEL

Low; uses a smartphone mainly
for calls and messaging



NEEDS

Easy-to-use app with clear
explanations, reminders, and
accessible health advice

Patient Persona Fatheya 75



KEY PAIN POINTS

1. Cannot understand medical scans or lab reports
2. Struggles to follow complex medicine instructions
3. Feels lost in crowded clinics
4. Needs guidance for post-surgery recovery and nutrition



OCCUPATION

A junior doctor at a rural clinic



TECH LEVEL

moderate; comfortable with apps, but busy with patients



NEEDS

AI-assisted support for early detection, structured explanations to portray results to patients

Doctor Persona Ahmed 23



KEY PAIN POINTS

1. High patient load with limited time per case
2. Difficulty explaining lab results to patients
3. Risk of inconsistent early diagnosis due to inexperience

PROBLEM EXPLORATION

1

Patient Related Issues



2

Doctor Related Issues



3

Pharmacies Related Issues



PROBLEM EXPLORATION

1

Doctor Related Issues



1



Difficulty understanding
medical scans/results

2



Confusion about medications &
side effects

3



Risk of misdiagnosis
or delayed diagnosis

PROBLEM EXPLORATION

1

Doctor Related Issues



4



Elderly or low-tech
patients struggle with
apps

5



Lack of guidance for
post-surgery recovery and nutrition

6



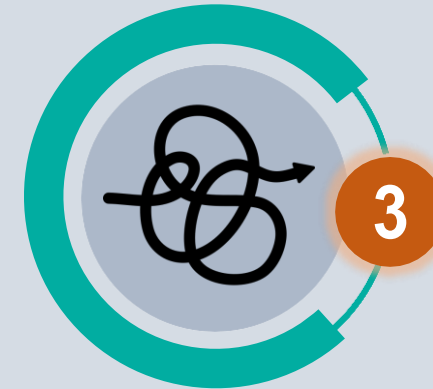
Long waiting times
in clinics

PROBLEM EXPLORATION

2

Doctor Related Issues

Need of support
for early
diagnosis



Complex lab
results
explaining



Risk of
inexperience

Overloaded
cases



PROBLEM EXPLORATION

3

Pharmacies Related Issues

1

Handwriting is often unclear

2

Risk of wrong drugs/dosages

3

Need for instructions clarity

4

AI-assisted alerts can help reduce human errors



PROPOSED WORK

Scan Analysis
(X-Rays , CT, MRI)



Lab test
interpretation



Simple explanation
of medical terms



Medical Intelligence
Everything the AI
analyses or explains

Personalized
diet plans



Nutrition &
health guidance



PROPOSED WORK

**Search for
doctors, clinics,
and pharmacies**



2

**Optional “case
summary” for
doctors**



**Healthcare Interaction
(Everything connecting
patients with real
healthcare providers)**

**View doctor profiles
(specialty, location,
contact)**



**Smart
Redirection to
trusted booking
platforms**



PROPOSED WORK

**Post-surgery
recovery guidance**



**Easy interface
for elderly**



Medicine reminders



**Patient Services
(Everything made for
the patient's daily use)**

3

**Medicine
Knowledge**



**Upload
scans/labs**



**Dual-language
support**



AI TECHNOLOGY

